Unit II: Partial Differentiation

Definition

Let $z = f(x, y)$ be a real-valued function of two independent variables x and y.

The partial derivative of $z = f(x, y)$ with respect to x measures the rate of change of f along the x-direction while keeping y fixed as a constant. It is denoted by $\frac{\partial z}{\partial x}$ or z_x, and is formally defined as:

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

provided this limit exists.

Similarly, the partial derivative of $f(x, y)$ with respect to y measures the rate of change along the y-direction while holding x constant. It is denoted by $\frac{\partial z}{\partial y}$ or z_y, and is defined as:

$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

provided the limit exists.

Examples

Question: Find the values of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point $(4, -5)$ for the function:

$$f(x, y) = x^2 + 3xy + y - 1$$

Solution:

Differentiating $f(x, y)$ partially with respect to x (treating y as constant):

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + 3xy + y - 1) = 2x + 3y$$

Evaluating at the point $(x, y) = (4, -5)$:

$$\left(\frac{\partial f}{\partial x}\right)_{(4,-5)} = 2(4) + 3(-5) = 8 - 15 = -7$$

Differentiating $f(x, y)$ partially with respect to y (treating x as constant):

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2 + 3xy + y - 1) = 3x + 1$$

Evaluating at the point $(x, y) = (4, -5)$:

$$\left(\frac{\partial f}{\partial y}\right)_{(4,-5)} = 3(4) + 1 = 13$$

Question: Find the first-order partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for each of the following functions:

1. $f(x,y) = x + y$
2. $f(x,y) = xy$
3. $f(x,y) = \frac{x}{y}$
4. $f(x,y) = \frac{x}{x^2+y^2}$
5. $f(x,y) = \frac{1}{x+y}$
6. $f(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$
7. $f(x,y) = e^{xy}$
8. $f(x,y) = e^{xy}\ln(y)$
9. $f(x,y) = x^y$
10. $f(x,y) = y\sin(xy)$

Solutions to the Problems

No.	Function $f(x,y)$	$\frac{\partial f}{\partial x}$	$\frac{\partial f}{\partial y}$
1	$x + y$	1	1
2	xy	y	x
3	$\frac{x}{y}$	$\frac{1}{y}$	$-\frac{x}{y^2}$
4	$\frac{x}{x^2+y^2}$	$\frac{y^2-x^2}{(x^2+y^2)^2}$	$-\frac{2xy}{(x^2+y^2)^2}$
5	$\frac{1}{x+y}$	$-\frac{1}{(x+y)^2}$	$-\frac{1}{(x+y)^2}$
6	$\tan^{-1}\left(\frac{y}{x}\right)$	$-\frac{y}{x^2+y^2}$	$\frac{x}{x^2+y^2}$
7	e^{xy}	ye^{xy}	xe^{xy}
8	$e^{xy}\ln(y)$	$ye^{xy}\ln(y)$	$e^{xy}\left(x\ln(y)+\frac{1}{y}\right)$
9	x^y	yx^{y-1}	$x^y\ln(x)$
10	$y\sin(xy)$	$y^2\cos(xy)$	$\sin(xy) + xy\cos(xy)$

Second-Order Partial Derivatives & Notations

For a function $z = f(x,y)$, taking partial derivatives repeatedly yields higher-order partial derivatives:

- Second order with respect to x:

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

- Mixed partial derivative (y then x):

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\,\partial y} = f_{xy}$$

- Mixed partial derivative (x then y):

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x} = f_{yx}$$

- Second order with respect to y:

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

Symmetry of Mixed Partials (Clairaut's / Schwarz's Theorem): If the partial derivatives are continuous in a region, the order of differentiation is commutative:

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}$$

Standard Notations

In differential geometry and partial differential equations, Monge's standard letter designations for partial derivatives of $z = f(x, y)$ are defined as:

Derivative Expression	Monge Symbol	Definition
$\frac{\partial z}{\partial x}$	p	First-order partial derivative wrt x
$\frac{\partial z}{\partial y}$	q	First-order partial derivative wrt y
$\frac{\partial^2 z}{\partial x^2}$	r	Second-order partial derivative wrt x
$\frac{\partial^2 z}{\partial x\,\partial y}$	s	Mixed second-order partial derivative
$\frac{\partial^2 z}{\partial y^2}$	t	Second-order partial derivative wrt y

Problem 1

Find the first and second order partial derivatives of $z = x^3 + y^3 - 3axy$.

Solution:

First-order partial derivatives:

$$\frac{\partial z}{\partial x} = 3x^2 - 3ay$$

$$\frac{\partial z}{\partial y} = 3y^2 - 3ax$$

Second-order partial derivatives:

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}(3x^2 - 3ay) = 6x$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}(3y^2 - 3ax) = 6y$$

$$s = \frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial}{\partial x}(3y^2 - 3ax) = -3a$$

Problem 2

If $u = x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$, show that $\frac{\partial^2 u}{\partial x\,\partial y} = \frac{x^2-y^2}{x^2+y^2}$ and verify that $\frac{\partial^2 u}{\partial x\,\partial y} = \frac{\partial^2 u}{\partial y\,\partial x}$.

Solution:

Differentiating u with respect to x:

$$\frac{\partial u}{\partial x} = 2x\tan^{-1}\left(\frac{y}{x}\right) + x^2 \cdot \frac{-\frac{y}{x^2}}{1+\left(\frac{y}{x}\right)^2} - y^2 \cdot \frac{\frac{1}{y}}{1+\left(\frac{x}{y}\right)^2}$$

Simplify each term:

$$x^2 \cdot \frac{-\frac{y}{x^2}}{\frac{x^2+y^2}{x^2}} = -\frac{x^2 y}{x^2+y^2}$$

$$y^2 \cdot \frac{\frac{1}{y}}{\frac{x^2+y^2}{y^2}} = \frac{y^3}{x^2+y^2}$$

Combining these terms:

$$\frac{\partial u}{\partial x} = 2x\tan^{-1}\left(\frac{y}{x}\right) - \frac{y(x^2+y^2)}{x^2+y^2} = 2x\tan^{-1}\left(\frac{y}{x}\right) - y$$

Now differentiate $\frac{\partial u}{\partial x}$ with respect to y:

$$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left[2x\tan^{-1}\left(\frac{y}{x}\right) - y\right] = 2x \cdot \frac{\frac{1}{x}}{1+\left(\frac{y}{x}\right)^2} - 1 = \frac{2x^2}{x^2+y^2} - 1 = \frac{x^2-y^2}{x^2+y^2}$$

By exact symmetry, differentiating first with respect to y and then x gives:

$$\frac{\partial^2 u}{\partial x\,\partial y} = \frac{x^2-y^2}{x^2+y^2}$$

Thus, $\frac{\partial^2 u}{\partial x\,\partial y} = \frac{\partial^2 u}{\partial y\,\partial x} = \frac{x^2-y^2}{x^2+y^2}$.

Problem 3

If $u = \frac{y}{x} + \frac{z}{x}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

Solution:

Rewrite u as $u = x^{-1}(y+z)$.

Compute the partial derivatives:

$$\frac{\partial u}{\partial x} = -x^{-2}(y+z) = -\frac{y+z}{x^2}$$

$$\frac{\partial u}{\partial y} = x^{-1} = \frac{1}{x}$$

$$\frac{\partial u}{\partial z} = x^{-1} = \frac{1}{x}$$

Substitute into the expression:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = x\left(-\frac{y+z}{x^2}\right) + y\left(\frac{1}{x}\right) + z\left(\frac{1}{x}\right)$$

$$= -\frac{y+z}{x} + \frac{y+z}{x} = 0$$

Problem 4

If $z(x+y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.

Solution:

Express z as $z = \frac{x^2+y^2}{x+y}$.

Differentiating z with respect to x:

$$\frac{\partial z}{\partial x} = \frac{(x+y)(2x) - (x^2+y^2)(1)}{(x+y)^2} = \frac{x^2+2xy-y^2}{(x+y)^2}$$

Differentiating z with respect to y:

$$\frac{\partial z}{\partial y} = \frac{(x+y)(2y) - (x^2+y^2)(1)}{(x+y)^2} = \frac{y^2+2xy-x^2}{(x+y)^2}$$

Compute the difference:

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = \frac{2x^2-2y^2}{(x+y)^2} = \frac{2(x-y)(x+y)}{(x+y)^2} = \frac{2(x-y)}{x+y}$$

Squaring both sides:

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = \frac{4(x-y)^2}{(x+y)^2}$$

Now compute the sum of the partial derivatives:

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{4xy}{(x+y)^2}$$

Evaluate the right-hand side:

$$4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = 4\left(1 - \frac{4xy}{(x+y)^2}\right) = 4\left(\frac{(x+y)^2 - 4xy}{(x+y)^2}\right) = \frac{4(x-y)^2}{(x+y)^2}$$

Both sides are equal, establishing the identity.

Problem 5

If $u = e^x \sin(y)$, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Solution:

Compute first and second derivatives:

$$\frac{\partial u}{\partial x} = e^x \sin(y), \quad \frac{\partial^2 u}{\partial x^2} = e^x \sin(y)$$

$$\frac{\partial u}{\partial y} = e^x \cos(y), \quad \frac{\partial^2 u}{\partial y^2} = -e^x \sin(y)$$

Adding the second-order partial derivatives:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin(y) - e^x \sin(y) = 0$$

Problem 6

If $z = \log(e^x + e^y)$, show that $rt - s^2 = 0$.

Solution:

First-order partial derivatives:

$$p = \frac{\partial z}{\partial x} = \frac{e^x}{e^x + e^y}$$

$$q = \frac{\partial z}{\partial y} = \frac{e^y}{e^x + e^y}$$

Second-order partial derivatives:

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{(e^x + e^y)e^x - e^x(e^x)}{(e^x + e^y)^2} = \frac{e^{x+y}}{(e^x + e^y)^2}$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{(e^x + e^y)e^y - e^y(e^y)}{(e^x + e^y)^2} = \frac{e^{x+y}}{(e^x + e^y)^2}$$

$$s = \frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{e^y}{e^x + e^y}\right) = -\frac{e^y e^x}{(e^x + e^y)^2} = -\frac{e^{x+y}}{(e^x + e^y)^2}$$

Evaluating $rt - s^2$:

$$rt - s^2 = \left(\frac{e^{x+y}}{(e^x + e^y)^2}\right)\left(\frac{e^{x+y}}{(e^x + e^y)^2}\right) - \left(-\frac{e^{x+y}}{(e^x + e^y)^2}\right)^2 = 0$$

Problem 7

If $u = (x^2 + y^2 + z^2)^{-1/2} = \frac{1}{\sqrt{x^2+y^2+z^2}}$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.

Solution:

First derivative with respect to x:

$$\frac{\partial u}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2x) = -x(x^2 + y^2 + z^2)^{-3/2}$$

Second derivative with respect to x:

$$\frac{\partial^2 u}{\partial x^2} = -(x^2 + y^2 + z^2)^{-3/2} - x\left(-\frac{3}{2}\right)(x^2 + y^2 + z^2)^{-5/2}(2x)$$

$$= -(x^2 + y^2 + z^2)^{-3/2} + 3x^2(x^2 + y^2 + z^2)^{-5/2}$$

By symmetry for y and z:

$$\frac{\partial^2 u}{\partial y^2} = -(x^2+y^2+z^2)^{-3/2} + 3y^2(x^2+y^2+z^2)^{-5/2}$$

$$\frac{\partial^2 u}{\partial z^2} = -(x^2+y^2+z^2)^{-3/2} + 3z^2(x^2+y^2+z^2)^{-5/2}$$

Summing the three terms:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -3(x^2+y^2+z^2)^{-3/2} + 3(x^2+y^2+z^2)(x^2+y^2+z^2)^{-5/2}$$

$$= -3(x^2+y^2+z^2)^{-3/2} + 3(x^2+y^2+z^2)^{-3/2} = 0$$

Problem 8

If $u = \log(x^3+y^3+z^3-3xyz)$, show that:

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) u = \frac{3}{x+y+z}$$

and:

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$$

Solution:

Using the algebraic identity $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$:

$$\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3y^2-3zx}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

Summing the three partial derivatives:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)} = \frac{3}{x+y+z}$$

Now applying the operator $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)$ to $\frac{3}{x+y+z}$:

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left(\frac{3}{x+y+z}\right)$$

$$= 3 \cdot \left(-\frac{1}{(x+y+z)^2} - \frac{1}{(x+y+z)^2} - \frac{1}{(x+y+z)^2} \right) = -\frac{9}{(x+y+z)^2}$$

Total Derivative

1. Total Derivative for Function of Two Variables:

If $u = f(x, y)$ where $x = \phi(t)$ and $y = \psi(t)$, then the total derivative of u with respect to t is:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

2. Total Derivative for Function of Three Variables:

If $u = f(x, y, z)$ where x, y, z are all functions of t, then:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$$

3. Derivative of an Implicit Function:

If $f(x, y) = c$ is an implicit function of x and y, then:

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \; where \; \frac{\partial f}{\partial y} \neq 0$$

Solved Problems

Problem 1

Question: Given $u = sin(\frac{x}{y})$, where $x = e^t$ and $y = t^2$, find $\frac{du}{dt}$ using the total derivative formula and verify the result by direct substitution.

Solution:

Method 1: Using Chain Rule (Total Derivative)

Given $u = sin(\frac{x}{y})$, $x = e^t$, and $y = t^2$.

1. Compute partial derivatives of u:

$$\frac{\partial u}{\partial x} = cos(\frac{x}{y}) \cdot \frac{\partial}{\partial x}(\frac{x}{y}) = \frac{1}{y}cos(\frac{x}{y})$$

$$\frac{\partial u}{\partial y} = cos(\frac{x}{y}) \cdot \frac{\partial}{\partial y}(\frac{x}{y}) = -\frac{x}{y^2}cos(\frac{x}{y})$$

2. Compute derivatives of x and y with respect to t:

$$\frac{dx}{dt} = e^t, \qquad \frac{dy}{dt} = 2t$$

3. Apply total derivative formula:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

$$\frac{du}{dt} = [\frac{1}{y}cos(\frac{x}{y})]e^t + [-\frac{x}{y^2}cos(\frac{x}{y})](2t)$$

$$\frac{du}{dt} = cos(\frac{x}{y})[\frac{e^t}{y} - \frac{2xt}{y^2}]$$

Substituting $x = e^t$ and $y = t^2$:

$$\frac{du}{dt} = cos(\frac{e^t}{t^2})[\frac{e^t}{t^2} - \frac{2t\ e^{\ t}}{t^4}]$$

$$\frac{du}{dt} = cos(\frac{e^t}{t^2}) \cdot \frac{e^t}{t^4}(t^2 - 2t)$$

$$\frac{du}{dt} = \frac{e^t(t - 2)}{t^3}cos(\frac{e^t}{t^2})$$

Verification by Direct Substitution:

Substitute x and y directly into u:

$$u = sin(\frac{e^t}{t^2})$$

Differentiating directly with respect to t using Quotient Rule:

$$\frac{du}{dt} = cos(\frac{e^t}{t^2}) \cdot \frac{d}{dt}(\frac{e^t}{t^2})$$

$$\frac{du}{dt} = cos(\frac{e^t}{t^2})[\frac{t^2e^t - 2te^t}{t^4}]$$

$$\frac{du}{dt} = \frac{e^t(t - 2)}{t^3}cos(\frac{e^t}{t^2})$$

Hence, the result is verified.

Problem 2

Question: If x is increasing at the rate of 2 cm/sec at the instant when $x = 3\ cm$ and $y = 1\ cm$, at what rate must y be changing in order that the function $u = 2xy - 3x^2y$ shall be neither increasing nor decreasing?

Solution:

Let $u = 2xy - 3x^2y$. For u to be neither increasing nor decreasing, its total rate of change must be zero, i.e., $\frac{du}{dt} = 0$.

1. Total derivative formula:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

2. Compute partial derivatives:

$$\frac{\partial u}{\partial x} = 2y - 6xy$$

$$\frac{\partial u}{\partial y} = 2x - 3x^2$$

3. Substitute values at $x = 3$ and $y = 1$:

$$\frac{\partial u}{\partial x} = 2(1) - 6(3)(1) = 2 - 18 = -16$$

$$\frac{\partial u}{\partial y} = 2(3) - 3(3^2) = 6 - 27 = -21$$

4. Substitute $\frac{dx}{dt} = 2$ and $\frac{du}{dt} = 0$ into the total derivative equation:

$$0 = (-16)(2) + (-21)(\frac{dy}{dt})$$

$$0 = -32 - 21\frac{dy}{dt}$$

$$\frac{dy}{dt} = -\frac{32}{21}\ cm/sec$$

Conclusion: Thus, y is decreasing at the rate of $\frac{32}{21}\ cm/sec$.

Problem 3

Question: The altitude of a right circular cone is 15 cm and is increasing at 0.2 cm/sec. The radius of the base is 10 cm and is decreasing at 0.3 cm/sec. How fast is the volume changing?

Solution:

The volume V of a right circular cone with radius r and height h is given by:

$$V = \frac{1}{3}\pi r^2 h$$

1. The rate of change of volume with respect to time t is:

$$\frac{dV}{dt} = \frac{\partial V}{\partial r}\frac{dr}{dt} + \frac{\partial V}{\partial h}\frac{dh}{dt}$$

2. Compute partial derivatives:

$$\frac{\partial V}{\partial r} = \frac{1}{3}\pi h(2r) = \frac{2}{3}\pi r h$$

$$\frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2$$

3. Given parameters: $h = 15\ cm$, $\frac{dh}{dt} = 0.2\ cm/sec$, $r = 10\ cm$, and $\frac{dr}{dt} = -0.3\ cm/sec$ (negative since decreasing).

4. Calculate numerical values of partial derivatives:

$$\frac{\partial V}{\partial r} = \frac{2}{3}\pi(10)(15) = 100\pi$$

$$\frac{\partial V}{\partial h} = \frac{1}{3}\pi(10^2) = \frac{100}{3}\pi$$

5. Substitute into total derivative:

$$\frac{dV}{dt} = (100\pi)(-0.3) + (\frac{100}{3}\pi)(0.2)$$

$$\frac{dV}{dt} = 100\pi[-0.3 + \frac{0.2}{3}] = 100\pi[\frac{-0.9 + 0.2}{3}]$$

$$\frac{dV}{dt} = 100\pi[\frac{-0.7}{3}] = -\frac{70}{3}\pi\ cm^3/sec$$

Conclusion: The volume of the cone is decreasing at the rate of $\frac{70}{3}\pi\ cm^3/sec \approx 73.30\ cm^3/sec$.

Problem 4

Question: If $u = sin^{-1}(x - y)$, where $x = 3t$ and $y = 4t^3$, show that:

$$\frac{du}{dt} = \frac{3}{\sqrt{1 - t^2}}$$

Solution:

Method 1: Using Total Derivative Formula

Given $u = sin^{-1}(x - y), x = 3t, y = 4t^3$.

1. Partial derivatives:

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1 - (x - y)^2}}$$

$$\frac{\partial u}{\partial y} = \frac{-1}{\sqrt{1 - (x - y)^2}}$$

2. Derivatives with respect to t:

$$\frac{dx}{dt} = 3, \qquad \frac{dy}{dt} = 12t^2$$

3. Substitute into total derivative:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

$$\frac{du}{dt} = \frac{3}{\sqrt{1 - (x - y)^2}} - \frac{12t^2}{\sqrt{1 - (x - y)^2}}$$

$$\frac{du}{dt} = \frac{3(1 - 4t^2)}{\sqrt{1 - (3t - 4t^3)^2}}$$

Simplifying denominator: $1 - (3t - 4t^3)^2 = 1 - 9t^2 + 24t^4 - 16t^6$ which factors cleanly as $(1 - 4t^2)^2(1 - t^2)$.

$$\frac{du}{dt} = \frac{3(1 - 4t^2)}{(1 - 4t^2)\sqrt{1 - t^2}} = \frac{3}{\sqrt{1 - t^2}}$$

Method 2: By Direct Substitution & Trigonometric Identity

Substitute x and y directly into u:

$$u = sin^{-1}(3t - 4t^3)$$

Recall the trigonometric identity: $sin(3\theta) = 3sin\theta - 4sin^3\theta$. Setting $t = sin\theta$ gives:

$$u = sin^{-1}(sin(3\theta)) = 3\theta = 3sin^{-1}(t)$$

Differentiating with respect to t:

$$\frac{du}{dt} = \frac{3}{\sqrt{1 - t^2}}$$

Hence proved.

Problem 5

Question: If $u = x\,log(xy)$ where $x^3 + y^3 + 3xy = 1$, find $\frac{du}{dx}$.

Solution:

Here u is a function of x and y, where y is an implicit function of x. The total derivative with respect to x is:

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$$

1. Compute partial derivatives of $u = x\,log(xy)$:

$$\frac{\partial u}{\partial x} = log(xy) + x \cdot \frac{1}{xy} \cdot y = log(xy) + 1$$

$$\frac{\partial u}{\partial y} = x \cdot \frac{1}{xy} \cdot x = \frac{x^2}{y}$$

2. Find $\frac{dy}{dx}$ using the implicit function formula:

Let $f(x, y) = x^3 + y^3 + 3xy - 1 = 0$.

$$\frac{\partial f}{\partial x} = 3x^2 + 3y$$

$$\frac{\partial f}{\partial y} = 3y^2 + 3x$$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{3x^2 + 3y}{3y^2 + 3x} = -\frac{x^2 + y}{y^2 + x}$$

3. Substitute into the total derivative expression:

$$\frac{du}{dx} = [log(xy) + 1] + [\frac{x^2}{y}] \cdot [-\frac{x^2 + y}{y^2 + x}]$$

$$\frac{du}{dx} = log(xy) + 1 - \frac{x^2(x^2 + y)}{y(y^2 + x)}$$

Problem 6

Question: If $\phi(x, y, z) = 0$, show that:

$$\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = -1$$

Solution:

$$\left(\frac{\partial y}{\partial z}\right)_x = -\frac{\frac{\partial \phi}{\partial z}}{\frac{\partial \phi}{\partial y}}, \quad \left(\frac{\partial z}{\partial x}\right)_y = -\frac{\frac{\partial \phi}{\partial x}}{\frac{\partial \phi}{\partial z}}, \quad \left(\frac{\partial x}{\partial y}\right)_z = -\frac{\frac{\partial \phi}{\partial y}}{\frac{\partial \phi}{\partial x}}$$

Multiplying all three partial derivatives together gives $\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = -1$.

Chain Rule in Partial Differentiation

1. Chain Rule for Function of Two Intermediate Variables:

If $u = f(r, s)$ where $r = \phi(x, y)$ and $s = \psi(x, y)$, then the partial derivatives of u are:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x}$$
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y}$$

2. Chain Rule for Function of Three Intermediate Variables:

If $u = f(r, s, t)$ where $r = \phi(x, y, z)$, $s = \psi(x, y, z)$, and $t = \chi(x, y, z)$, then:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$$
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$$
$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial z}$$

Solved Problems

Problem 1

Question: Find $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ for: (i) $u = f(x + y, x - y)$ (ii) $u = f(x + 2y, 2x - y)$

Solution (i):

Let $r = x + y$ and $s = x - y$. Then $u = f(r, s)$.

By chain rule:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} = \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(1) = \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} = \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial s}$$

Adding the two equations:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2\frac{\partial u}{\partial r} = 2\frac{\partial f}{\partial r}$$

Solution (ii):

Let $r = x + 2y$ and $s = 2x - y$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(2) = \frac{\partial u}{\partial r} + 2\frac{\partial u}{\partial s}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}(2) + \frac{\partial u}{\partial s}(-1) = 2\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s}$$

Adding the two equations:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$

Problem 2

Question: If $u = f(x^2 + 2yz,\ y^2 + 2zx,\ z^2 + 2xy)$, show that:

$$(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$$

Solution:

Let $r = x^2 + 2yz, s = y^2 + 2zx, t = z^2 + 2xy$.

1. Compute partial derivatives of r, s, t with respect to x, y, z:

$$\frac{\partial r}{\partial x} = 2x, \quad \frac{\partial s}{\partial x} = 2z, \quad \frac{\partial t}{\partial x} = 2y$$

$$\frac{\partial r}{\partial y} = 2z, \quad \frac{\partial s}{\partial y} = 2y, \quad \frac{\partial t}{\partial y} = 2x$$

$$\frac{\partial r}{\partial z} = 2y, \quad \frac{\partial s}{\partial z} = 2x, \quad \frac{\partial t}{\partial z} = 2z$$

2. Apply chain rule for $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$:

$$\frac{\partial u}{\partial x} = 2x\frac{\partial u}{\partial r} + 2z\frac{\partial u}{\partial s} + 2y\frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial y} = 2z\frac{\partial u}{\partial r} + 2y\frac{\partial u}{\partial s} + 2x\frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial z} = 2y\frac{\partial u}{\partial r} + 2x\frac{\partial u}{\partial s} + 2z\frac{\partial u}{\partial t}$$

3. Multiplying by $(y^2 - zx), (x^2 - yz), (z^2 - xy)$ respectively and summing terms gives coefficients of $\frac{\partial u}{\partial r}, \frac{\partial u}{\partial s}, \frac{\partial u}{\partial t}$ equal to zero.

Hence proved.

Problem 3

Question: If $u = f(x - y, y - z, z - x)$, prove that:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

Solution:

Let $r = x - y, s = y - z, t = z - x$.

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t}(-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}(-1) + \frac{\partial u}{\partial s}(1) + \frac{\partial u}{\partial t}(0) = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}(-1) + \frac{\partial u}{\partial t}(1) = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t}$$

Adding all three equations:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (\frac{\partial u}{\partial r} - \frac{\partial u}{\partial t}) + (-\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}) + (-\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t}) = 0$$

Hence proved.

Problem 4

Question: If $z = f(x, y)$, where $x = e^u \cos v$ and $y = e^u \sin v$, prove that:

$$(i)\ x\frac{\partial z}{\partial v} + y\frac{\partial z}{\partial u} = e^{2u}\frac{\partial z}{\partial y}$$

$$(ii)\ \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$$

Solution:

Compute partial derivatives of x and y with respect to u and v:

$$\frac{\partial x}{\partial u} = e^u \cos v = x, \qquad \frac{\partial x}{\partial v} = -e^u \sin v = -y$$

$$\frac{\partial y}{\partial u} = e^u \sin v = y, \qquad \frac{\partial y}{\partial v} = e^u \cos v = x$$

By chain rule:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}(x) + \frac{\partial z}{\partial y}(y) = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}(-y) + \frac{\partial z}{\partial y}(x) = -y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y}$$

For part (i): Substitute into $x\frac{\partial z}{\partial v} + y\frac{\partial z}{\partial u}$:

$$x(-y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y}) + y(x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y})$$

$$= (x^2 + y^2)\frac{\partial z}{\partial y} = (e^{2u}cos^2 v + e^{2u}sin^2 v)\frac{\partial z}{\partial y} = e^{2u}\frac{\partial z}{\partial y}$$

For part (ii): Square and add $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$:

$$\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = (x^2 + y^2)\left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right]$$

$$= e^{2u}\left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right]$$

Dividing by e^{2u} completes the proof.

Problem 5

Question: If $z = f(x, y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, prove that:

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$$

Solution:

Compute partial derivatives of x and y:

$$\frac{\partial x}{\partial u} = e^u, \quad \frac{\partial x}{\partial v} = -e^{-v}$$

$$\frac{\partial y}{\partial u} = -e^{-u}, \quad \frac{\partial y}{\partial v} = -e^v$$

Applying chain rule:

$$\frac{\partial z}{\partial u} = e^u\frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial v} = -e^{-v}\frac{\partial z}{\partial x} - e^v\frac{\partial z}{\partial y}$$

Subtracting $\frac{\partial z}{\partial v}$ from $\frac{\partial z}{\partial u}$:

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = (e^u + e^{-v})\frac{\partial z}{\partial x} - (e^{-u} - e^v)\frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$$

Hence proved.

Problem 6

Question: If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that:

$$\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$$

Solution:

Let $r = 2x - 3y, s = 3y - 4z, t = 4z - 2x$.

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial r} - 2\frac{\partial u}{\partial t} \Rightarrow \frac{1}{2}\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial y} = -3\frac{\partial u}{\partial r} + 3\frac{\partial u}{\partial s} \Rightarrow \frac{1}{3}\frac{\partial u}{\partial y} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$

$$\frac{\partial u}{\partial z} = -4\frac{\partial u}{\partial s} + 4\frac{\partial u}{\partial t} \Rightarrow \frac{1}{4}\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t}$$

Adding the three equations:

$$\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$$

Hence proved.

Problem 7

Question: If $u = f(\frac{y-x}{xy}, \frac{z-x}{xz})$, show that:

$$x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$$

Solution:

Rewrite arguments as $r = \frac{1}{x} - \frac{1}{y}$ and $s = \frac{1}{x} - \frac{1}{z}$.

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}(-\frac{1}{x^2}) + \frac{\partial u}{\partial s}(-\frac{1}{x^2}) \Rightarrow x^2\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}(\frac{1}{y^2}) \Rightarrow y^2\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial s}(\frac{1}{z^2}) \Rightarrow z^2\frac{\partial u}{\partial z} = \frac{\partial u}{\partial s}$$

Summing the three expressions gives 0.

Hence proved.

Problem 8

Question: If $u = f(x, y)$ where $x = r\,cos\,\theta$ and $y = r\,sin\,\theta$, prove that:

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

Solution:

Apply chain rule for polar coordinates transformation:

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} cos\,\theta + \frac{\partial u}{\partial y} sin\,\theta$$

$$\frac{\partial u}{\partial \theta} = -r\,\frac{\partial u}{\partial x} sin\,\theta + r\,\frac{\partial u}{\partial y} cos\,\theta$$

Dividing $\frac{\partial u}{\partial \theta}$ by r, then squaring and adding both equations:

$$\left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 (cos^2\theta + sin^2\theta) + \left(\frac{\partial u}{\partial y}\right)^2 (sin^2\theta + cos^2\theta)$$

$$= \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2$$

Hence proved.